

Theory of Computation (CS2030) : Problem Set 1

Department of Computer Science and Engineering
IIT Hyderabad

Please read the following comment before you work on the problems.

This is a set of questions that you should solve to master the contents taught in the class. *You don't have to submit the solutions!* But you should still solve them to understand the material, and also because these problems or their variants have a devilish way of finding their way in exams/quizzes!

We suggest to try to solve them from scratch under exam conditions—by yourself, *without* your notes, *without* the internet, and if possible, even without a cheat-sheet.

If you find yourself getting stuck on a problem, try to figure out *why* you're stuck. Do you understand the problem statement? Are you stuck on choosing the right high-level approach, or are you stuck on the details? Are you solving the right subproblems?

Discussing problems with other people (in your study groups, with me, on Google Classroom, or with LLMs) and/or looking up old solutions can be *extremely* helpful, but **only after** you have (1) made a good-faith effort to solve the problem on your own, and (2) you have either a candidate solution or some idea about where you are getting stuck.

When you do discuss problems with other people, remember that your goal is not merely to “understand” the solution to any particular problem, but to become more comfortable with solving a certain type of problem on your own. Identify specific steps that you find problematic, read more about those steps, and focus your practice on those steps.

1. Show that the following languages are regular.

- (a) $\{w \in \{a, b\}^* \mid w \text{ has at exactly two } a\text{'s and at least two } b\text{'s}\}$
- (b) $\{w \in \{a, b\}^* \mid w \text{ has an even number of } a\text{'s and one or two } b\text{'s}\}$
- (c) $\{w \in \{a, b\}^* \mid w \text{ has an even number of } a\text{'s and each } a \text{ is followed by at least one } b\}$
- (d) $\{w \in \{a, b\}^* \mid w \text{ has an odd number of } a\text{'s and ends with a } b\}$
- (e) $\{w \in \{a, b\}^* \mid w \text{ has even length and an odd number of } a\text{'s}\}$
- (f) $\{w \in \{a, b\}^* \mid w \text{ does not contain the substring } baba\}$
- (g) $\{w \in \{a, b\}^* \mid w \text{ contains neither the substrings } ab \text{ nor } ba\}$
- (h) $\{w \in \{a, b\}^* \mid w \text{ is any string not in } a^* \cup b^*\}$
- (i) $\{w \in \{0, 1\}^* \mid w \text{ has length at least 3 and its third symbol is a 0}\}$
- (j) $\{w \in \{0, 1\}^* \mid w \text{ starts with 0 and has odd length, or starts with 1 and has even length}\}$
- (k) $\{w \in \{0, 1\}^* \mid \text{every odd position of } w \text{ is a 1}\}$

2. Prove that every NFA can be converted to an equivalent one that has a single accept state.
3. Let $D = \{w \mid w \text{ contains an even number of } a\text{'s and an odd number of } b\text{'s and does not contain the substring } ab\}$. Give a DFA that recognizes D and a regular expression that generates D .
4. Let F be the language of all strings over $\{0, 1\}$ that do not contain a pair of 1s that are separated by an odd number of symbols. Show that F is regular.
5. In certain programming languages, comments appear between delimiters such as $/\#$ and $\#/\#$. Let C be the language of all valid delimited comment strings. A member of C must begin with $/\#$ and end with $\#/\#$ but have no intervening $\#/\#$. For simplicity, we'll say that the comments themselves are written with only the symbols a and b ; hence the alphabet of C is $\Sigma = \{a, b, /, \#\}$. Give a DFA that recognizes C .
6. Let B be any language over the alphabet Σ . Define $B^+ := \bigcup_{n \geq 1} B^n$. Prove that $B = B^+$ iff $BB \subseteq B$.
7. For any string $w = w_1w_2 \cdots w_n$, the *reverse* of w , denoted $\text{rev}(w)$, is the string w in reverse order, $w_n \cdots w_2w_1$. For any language A , let $\text{rev}(A) = \{\text{rev}(w) \mid w \in A\}$. Show that if A is regular, then so is $\text{rev}(A)$.
8. Let

$$\Sigma_3 = \left\{ \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \dots, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}.$$

That is, Σ_3 contains all size 3 columns of 0s and 1s. A string of symbols in Σ_3 gives three rows of 0s and 1s. Consider each row to be a binary number and let $B = \{w \in \Sigma_3^* \mid \text{the bottom row of } w \text{ is the sum of the top two rows}\}$. For example,

$$\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \in B, \quad \text{but} \quad \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \notin B.$$

Show that B is regular.

9. An **all**-NFA M is a 5-tuple $(Q, \Sigma, \delta, q_0, F)$ that accepts $x \in \Sigma^*$ if *every* possible state that M could be in after reading input x is a state from F . Note, in contrast, that an ordinary NFA accepts a string if some state among these possible states is an accept state. Prove that **all**-NFAs recognize the class of regular languages.
10. A string x is said to be a *prefix* of string y if there exists another string $z \in \Sigma^*$ such that $y = xz$. Furthermore, x is a *proper prefix* if in addition $z \neq \epsilon$. For any language

$A \subseteq \Sigma^*$, define

$$\text{noprefix}(A) = \{w \in A \mid \text{no proper prefix of } w \text{ is a member of } A\}, \text{ and}$$

$$\text{noextend}(A) = \{w \in A \mid w \text{ is not the proper prefix of any string in } A\}.$$

Show that if A is regular, then so are $\text{noprefix}(A)$ and $\text{noextend}(A)$.

11. For languages $A, B \subseteq \Sigma^*$, let the *perfect shuffle* of A and B , denoted $\text{p-shuffle}(A, B)$, be the language

$$\{w \mid w = a_1b_1 \cdots a_kb_k, \text{ where } a_1 \cdots a_k \in A \text{ and } b_1 \cdots b_k \in B, \text{ each } a_i, b_i \in \Sigma\}.$$

Show that if A and B are regular, then $\text{p-shuffle}(A, B)$ is also regular.

12. For languages $A, B \subseteq \Sigma^*$, let the *shuffle* of A and B , denoted $\text{shuffle}(A, B)$, be the language

$$\{w \mid w = a_1b_1 \cdots a_kb_k, \text{ where } a_1 \cdots a_k \in A \text{ and } b_1 \cdots b_k \in B, \text{ each } a_i, b_i \in \Sigma^*\}.$$

Show that if A and B are regular, then $\text{shuffle}(A, B)$ is also regular.

13. Let $A/B = \{w \mid wx \in A \text{ for some } x \in B\}$. Show that if A is regular and B is any language then A/B is regular.
14. The *Hamming distance* between two bit strings x and y , denoted $H(x, y)$, is the number of places at which they differ. For example, $H(011, 110) = 2$. (If $|x| \neq |y|$, then their Hamming distance is infinite.) For any set $A \subseteq \{0, 1\}^*$, define $N_2(A) = \{x \mid \text{there exists } y \in A \text{ such that } H(x, y) \leq 2\}$. Show that if A is regular, then so is $N_2(A)$.
15. For any $A \subseteq \Sigma^*$, define

$$\text{half}(A) = \{y \mid \text{there exists } x \text{ such that } |x| = |y| \text{ and } xy \in A\}.$$

Show that if A is regular, then so is $\text{half}(A)$.

16. For any $A \subseteq \Sigma^*$, define

$$\text{first-thirds}(A) = \{x \mid \text{there exists } y \text{ such that } |y| = 2|x| \text{ and } xy \in A\}.$$

Show that if A is regular, then so is $\text{first-thirds}(A)$.

17. Fix Σ and let $b \in \Sigma$. For a set $A \subseteq \Sigma^*$, let $A' = \{xy \mid xby \in A\}$. That is, A' consists of all strings obtained from a string in A by deleting exactly one b .
- Show that if A is regular, then so is A' .